F79MA 2025-26
Assessed Project 1

Project description

In financial risk management, extreme events such as sudden market crashes, insurance claim spikes, or operational losses often follow heavy-tailed distributions rather than the standard normal assumption. These distributions place more probability on extreme values, making them critical for accurate modelling of risk exposure. A poor choice of estimator can underestimate such risks, leading to flawed decision-making and financial vulnerability. This project will help you explore whether the maximum likelihood estimator (MLE) is reliable in this context by combining both theoretical derivations and simulation studies in R. As part of a financial risk team, you are asked to consider whether a continuous distribution could provide a suitable model for certain heavy-tailed risks. The distribution under consideration is defined by the probability density function (pdf)

$$f(x;\theta) = \tfrac{1}{2}\exp(-|x-\theta|), \quad -\infty < x < \infty,\ \theta \in \mathbb{R},$$

and the cumulative distribution function (cdf)

$$F(x;\theta) = \begin{cases} \tfrac{1}{2}\exp(x-\theta), & x < \theta, \\ 1 - \tfrac{1}{2}\exp(-(x-\theta)), & x \geq \theta. \end{cases}$$

Here, θ is the location parameter of the distribution.

Your task is to write a report discussing whether the maximum likelihood estimator (MLE) $\hat{\theta}$ is appropriate for use in practice. This requires both theoretical analysis and simulation in R.

To support this, you are provided with the R script `F79MA_Assignment1.R` (available on Canvas). In this script, you must set the argument of `set.seed` equal to the last four digits of your Heriot-Watt student ID number. For example, if your student ID is `H12345678`, then your seed should be `5678`. **Failure to do so will result in 0 marks being awarded.**

This R script will then generate a parameter `My_theta` based on your Student ID and an $m \times n$ matrix `samples` consisting of $m = 1000$ simulations of samples of size $n = 100$ from

the given distribution with parameter `My_theta`, which you can use for your analysis. In other words, each row corresponds to one sample from the given distribution. Do not change any of the code above the line reading 'Please insert your R code after this line' except for the seed. **Failure to do so may result in 0 marks being awarded.**

You must submit, using the links on Canvas: (a) your written report, as a pdf file or as a Word document; and (b) an R script containing code that calculates required quantities and produces any plots you include. Before submitting, you must complete the Standard Declaration of Student Authorship quiz on Canvas.

Tasks

1. Derive the likelihood $L(\theta; \underline{x})$ and log-likelihood $l(\theta; \underline{x}) = \log(L(\theta; \underline{x}))$ for a sample $\underline{x} = (x_1, \ldots, x_n)$. Clearly state any assumptions made. [2 marks]

2. Derive expressions for the score function $U(\theta)$, the Fisher information $I(\theta)$, and the Cramér–Rao Lower Bound (CRLB) for a sample $\underline{x} = (x_1, \ldots, x_n)$. [2.5 marks]

   Hint: You may find it useful to use $\mathrm{E}[U(\theta)^2]$ when deriving the formula for $I(\theta)$.

3. Derive a formula for the maximum likelihood estimator (MLE) $\hat{\theta}$ based on a sample $\underline{x} = (x_1, \ldots, x_n)$. Show that the MLE corresponds to the sample median. Also, comment on whether the CRLB is attainable in this case. [2.5 marks]

4. Using R, calculate the maximum likelihood estimate $\hat{\theta}$ for the first simulated sample (the first row of your matrix `samples`). Save this as `Quantity1` in your code. [0.5 marks]

5. Calculate $\ell(\hat{\theta}; \underline{x})$ for the maximum likelihood estimate `Quantity1` (without ignoring constants) and call it `Quantity2`. Create an appropriate plot in R to demonstrate that `Quantity1` maximises the log-likelihood. [1 mark]

6. Using the asymptotic normality of the MLE, calculate in R from the first sample an approximate 90% confidence interval for θ. Save the lower limit as `Quantity3` and the upper limit as `Quantity4`. Ensure that your confidence interval does not use the value of `My_theta` as part of the calculation. [1 mark]

7. Using the asymptotic distribution of the likelihood-ratio (deviance) statistic, calculate in R from the first sample an approximate 90% confidence interval for θ. Save the lower limit as `Quantity5` and the upper limit as `Quantity6`.

   Comment on the two confidence intervals obtained from the asymptotic normality and likelihood-ratio (deviance) approaches. [1.5 mark]

8. Using the full matrix `samples`, estimate the bias, variance, and mean squared error (MSE) of the MLE. Save the bias as `Quantity7`, the variance as `Quantity8`, and the MSE as `Quantity9`. [1.5 marks]

9. In R, calculate the CRLB for a smooth unbiased estimator for a sample of size n and save it as `Quantity10`. [0.5 marks]

10. By modifying the given code, plot how the bias, variance, and MSE of the MLE vary as n increases. Comment on whether the MLE appears consistent and efficient. [3 marks]

The Report

Your findings should be presented in the form of a report for your line manager, who is also part of the statistical modelling team but has not worked directly on this problem.

Your report should:

- have a clear and logical structure; include an introduction and clearly stated conclusions that can be understood by any numerate scientist;
- include detail of your mathematical calculations so that your results could be reproduced by another statistician;
- include clearly labelled and correctly referenced tables and diagrams, as appropriate;
- include citation and referencing for any material (books, papers, websites etc) used. When possible, use reliable sources, produced by respected and well-known authors, published by recognised publishers and associated with well established government, academic, or educational institutions. Note that some webpages, YouTube videos, blog posts, and Wikipedia pages might include errors.

Your report should be produced using a word-processing program, such as Microsoft Word, or with LaTeX, and uploaded as a single file in PDF format. All equations should be produced digitally (typeset) by using an equation editor (we recommend to avoid scanned or handwritten equations). You may use the Microsoft Word template provided, but this is not compulsory.

The report has a **maximum page limit of ten (10) pages** (equivalent to 5 double-sided A4 sheets, 11-point font).

A total of **4 Marks** is available for these aspects of your report. This will be marked according to the rubric given in the Appendix.

[Total: 20 Marks]

Notes

- **Errors in code:** Five marks are for correctly carrying out calculations in R (i.e. calculating the correct values for the Quantities), and this will be assessed by running your R code and comparing the results with the verified correct output. You must therefore ensure that you have defined variables as above (i.e. **exactly the same naming convention with capital Q and no underscore between Quantity and the number**) and ensure your code runs without generating errors. **Failure to do so may result in 0 marks being awarded**. You can check your code runs correctly by running it through the file `TestSubmissionAssignment1.R`.

- **Use of libraries and good practice with code:** Please do not give references to directories on your own computer (e.g. specifying a directory with setwd) or make use of any R library aside from the 'ggplot2' library (you do not need to use this library). Your code should be commented appropriately to ease with checking.

- **Questions for the Lecturers**: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly. That way all students will see the same responses. The discussion board will be closed at 10:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time on Tuesday 21st October 2025.

- **Collaboration and Plagiarism**: Students are allowed to discuss the methods used with other students, but your submitted project must be **all your own work**. Particularly please note the following:

  - Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
  - Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course or worse.
  - Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
  - Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

More information on plagiarism can be found here

- **Submission**: The report and the code should be submitted through Canvas using the appropriate links. It should not be submitted by email or handed in as a paper copy. You should also complete the Student Declaration of Authorship quiz.
- **Deadline**: The deadline for submission is **10:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time** on **Wednesday 22nd October 2025** (Week 7); projects may be submitted early. Late projects will be marked in line with the University Coursework Policy, i.e. a standard deduction of 30% if submitted within 5 working days and 0 marks awarded if submitted later. You can, however, submit a mitigating circumstances applications, but these may only be applied at the end of the semester. No individual extensions are allowed.
- **Feedback**: It is expected that feedback on this assignment will be available by Wednesday 12th November 2025 (Week 10).
- **Assessment**: This coursework will contribute 20% to the final mark for the course.

Appendix: Rubric for marking of the report

The five marks available for the exposition of your report will be awarded according to the scale below:

Between 0 and 1 marks will be awarded for	• Lack of clear and logical structure • Conclusions missing or not suitable for a non-statistician • Statistical calculations and methodology not clearly set out for the reader • Tables and figures unclear, badly labelled or not correctly referred to • Sources used not clearly referenced
Between 2 and 3 marks will be awarded for	• Clear and logical structure including Introduction and Conclusion • Conclusions generally suitable for a non-statistician • Statistical calculations and methodology generally set out clearly for the reader • Tables and figures often clear and correctly referred to • Sources used clearly referenced
4 marks will be awarded for	• Clear and logical structure including Introduction and Conclusion • Conclusions suitable for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures clear, correctly referred to and easy to interpret • Sources used clearly and correctly referenced